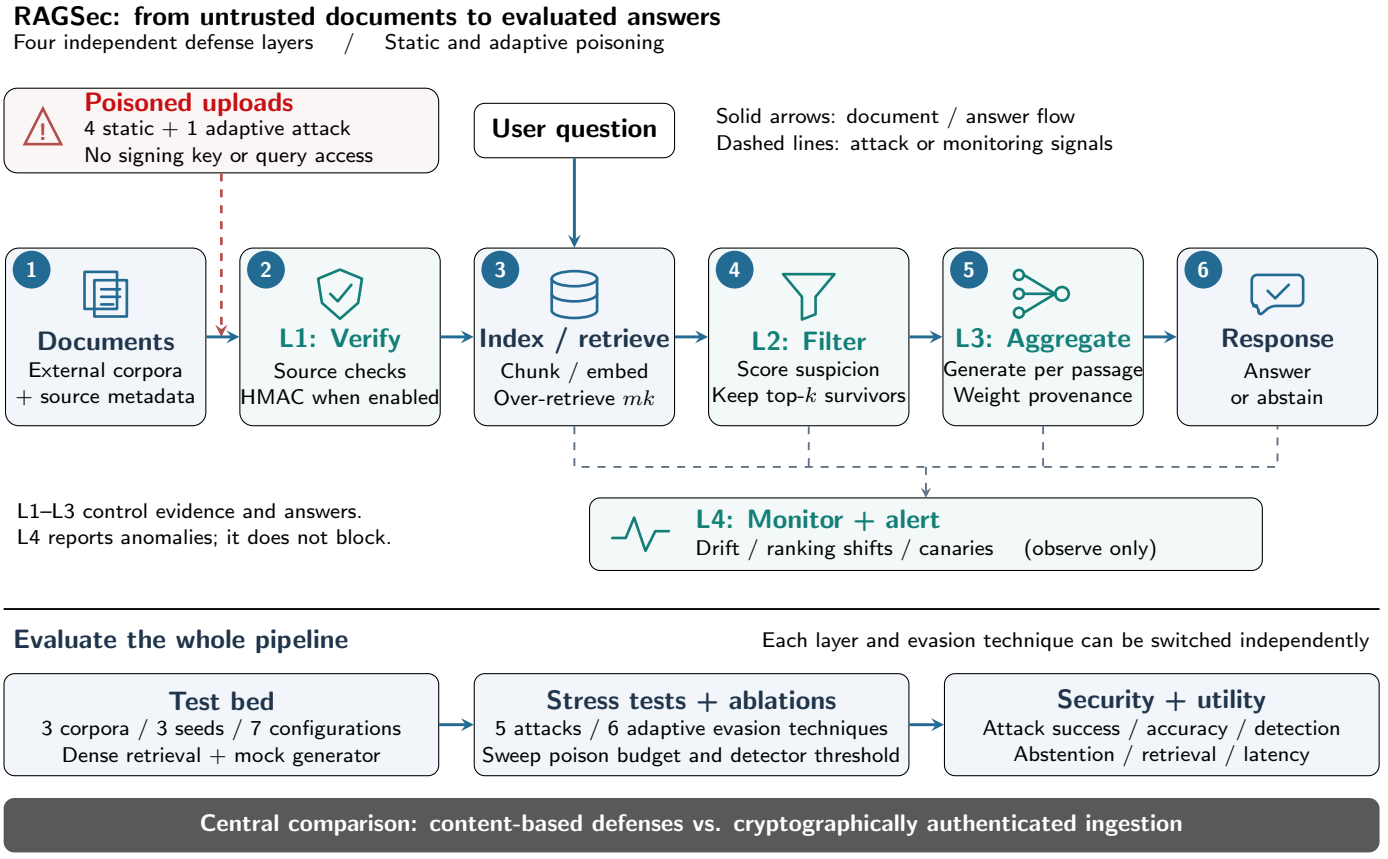
RAGSec: from untrusted documents to evaluated answers
Four independent defense layers / Static and adaptive poisoning
Poisoned uploads
4 static + 1 adaptive attack
No signing key or query access
User question
Solid arrows: document / answer flow
Dashed lines: attack or monitoring signals
1
Documents
External corpora
+ source metadata
2
L1: Verify
Source checks
HMAC when enabled
3
Index / retrieve
Chunk / embed
Over-retrieve mk
4
L2: Filter
Score suspicion
Keep top-k survivors
5
L3: Aggregate
Generate per passage
Weight provenance
6
Response
Answer
or abstain
L1–L3 control evidence and answers.
L4 reports anomalies; it does not block.
L4: Monitor + alert
Drift / ranking shifts / canaries (observe only)
Evaluate the whole pipeline
Each layer and evasion technique can be switched independently
Test bed
3 corpora / 3 seeds / 7 configurations
Dense retrieval + mock generator
Stress tests + ablations
5 attacks / 6 adaptive evasion techniques
Sweep poison budget and detector threshold
Security + utility
Attack success / accuracy / detection
Abstention / retrieval / latency
Central comparison: content-based defenses vs. cryptographically authenticated ingestion